Dr. Harun AKSAYA

Researcher · Software Developer · AI & NLP

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Google Scholar • github.com/haksaya • @harunaksaya

Research Interests

Artificial Intelligence · Natural Language Processing · Sentiment Analysis · Text Mining · Semantic Networks · Turkish Language Processing · Machine Learning · Educational Technology

Education

2016 – 2021 **Ph.D. in Informatics**
İstanbul University, Department of Informatics
GPA: 3.93/4.00 — High Honors

2013 – 2015 **M.Sc. in Computer Science**
Marmara University
GPA: 90.5/100 — High Honors

2011 – 2013 **Language School**
Middle East Technical University (METU)
High Honors

2006 – 2010 **B.Sc. in Computer Education and Instructional Technologies**
Marmara University
GPA: 85.93/100 — Graduated Top of the Department with Honors

Professional Experience

2012 – Present **Software Development Specialist**
Marmara University, İstanbul

2010 – 2012 **Software Development Specialist**
Middle East Technical University, Ankara

Teaching

2026 Spring **Lecturer — Text Mining**
İstanbul University, Department of Computer Science
Topics: text preprocessing, feature extraction, classification, clustering, sentiment analysis, NLP applications

Various terms

Instructor — Software and AI Specialization Certificate Programs

Arı Bilgi Eğitim Kurumları

Publications

Journal Articles

2026 Aksaya, H. & Gülseçen, S. **Aspect-Based Sentiment Analysis of Turkish School Reviews: A Comparative Study of Recurrent and Transformer Architectures**
Electronics, 15(21), 4949. DOI: 10.3390/electronics15214949

2026 Aksaya, H. & Gülseçen, S. **A Novel Lexicon-Based Approach for Sentiment Analysis in Turkish**
Applied Sciences, 16(13), 6612. DOI: 10.3390/app16136612

2015 Aksaya, H., Arşık, A., Elibol, M., Erdoğan Coşkun, A., Olgun, K.B., Özcan, M., & Reis, Z.A. **I Learn For My Hobby: How Much Are Aquarists Satisfied?**
The Turkish Online Journal of Educational Technology (TOJET), Vol. 2015 (Special Issue), pp. 311–324. Scopus.

2011 Esgin, E., Aksaya, H., Kırçalı, O., Direk, A., & Kılıç, M. **Determining the Factors of Interest to Digital Games and Comparing Them with Properties of Educational Games**
Gaziantep University Journal of Social Sciences, Vol. 10, No. 4, pp. 1291–1310. [20 Citations]

Thesis

2014 Aksaya, H. **An Investigation of Open Courseware Platforms from a Lifelong Learning Perspective**
M.Sc. Thesis, Atatürk Faculty of Education.

Book Chapters

2021 Bozbuğa, N., Gülseçen, S., Ak, M., & Aksaya, H. **Medical Informatics**
Book Chapter — Computer Science, Interdisciplinary Applications. DOI: 10.26650/B/ET07.2021.003

2021 Gülseçen, S. & Aksaya, H. **Clinical Decision Support Systems**
Book Chapter — Tıp Bilişimi (Medical Informatics), Istanbul University Press. DOI: 10.26650/B/ET07.2021.003.15

2012 Benzer, A. & Aksaya, H. (Eds.) **Bilgisayar Okuryazarlığı I-II (Computer Literacy I-II)**
Book editorship and chapter. Pegem Akademi Yayıncılık, 1st Edition. ISBN 978-605-364-357-9.

Conference Papers

2026 Aksaya, H., Olgun, K.B., & Elibol, M. **Türkçe Dil Modeli Geliştirme Alanındaki Çalışmaların Sistematik Değerlendirilmesi**
24. Akademik Bilişim Konferansı (AB'26), İstanbul Beykent University, Turkey (February 3–5, 2026). Full paper.

2025 Aksaya, H., Gülseçen, S., & Lepkova, N. **Comparison of Artificial Neural Networks with Dictionary-Based Approach in Aspect-Based Sentiment Analysis**
International Conference on Automatics and Informatics (ICAI'25), Varna, Bulgaria. IEEE Xplore / SCOPUS.

2020 Aksaya, H. & Gülseçen, S. **Developing a Text Mining Based System for Educational Institutions: Implementation of Machine Learning Techniques in Aspect-Based Sentiment Analysis Studies**
FutureLearning 2020: 8th International Conference on Future Learning and Informatics (FL2020), Istanbul University, October 20–22, 2020. Istanbul University Press.

2014 Arslan, A. & Aksaya, H. **A Scale Development Study for Evaluating Open Course Ware Platforms**
8th International Computer and Instructional Technologies Symposium (ICITS 2014), Trakya University, Edirne, Turkey.

Invited Talks & Presentations

2025 Aksaya, H. **From Data to Value | Artificial Intelligence as a Lever in Financial Systems**
XIII. Turkish Accounting Expertise Congress, Gaziantep, Turkey (December 5–7, 2025).

2025 Aksaya, H. **Artificial Intelligence as a Lever in Vocational Education and Industry**
XIII. International Vocational Schools Symposium (UMYOS 2025), Bursa Uludağ University, Turkey.

2021 Aksaya, H. **Instructor — 4th Artificial Intelligence Summer School (YAZSUM 2021)**
Online (September 1–3, 2021).

2020 Aksaya, H. **New Approaches in Natural Language Processing Studies**
Artificial Intelligence Summer School 2020, Sakarya University, Turkey.

Projects

Principal investigator of institutional information management system projects at Marmara University.

- **MİTTO - Patent Application System** (ongoing)
- **Mentor Marmara** (completed)

- **Proje Havuzu** (completed)
- **ÜNSİMER PBS** (completed)

Academic Service

Peer Reviewer — Journals

2026 **Reviewer**
Acta INFOLOGICA (ACIN)
Istanbul University, Department of Informatics. Peer-reviewed scientific journal (bilingual until March 2022; English only from March 2022). Published biannually.

2026 **Reviewer**
Beykent University Journal of Science and Engineering (BUJSE)
Istanbul Beykent University. International peer-reviewed journal in natural sciences, engineering, and architecture. Published biannually.

Peer Reviewer — Conferences

2026 **Reviewer and Session Chair**
24th Akademik Bilişim Conference (AB 2026)
İstanbul Beykent University, Turkey (February 3–5, 2026).

Jury & Committee

2025 **Jury Member**
Bilişim Yıldızları 2025 — National and Original Project Competition
Turkey Information Technology Association (TBD), Istanbul Branch.

Awards & Honors

2021 **High Honors — Ph.D.**
İstanbul University
GPA: 3.93 / 4.00

2015 **High Honors — M.Sc.**
Marmara University
Grade: 90.5 / 100

2013 **High Honors — Language School**
METU

2010 **Graduated Top of the Department with Honors — B.Sc.**
Marmara University
Grade: 85.93 / 100

Last updated: November 2026